

Week 9 Tutorial – Time Series

Question 1

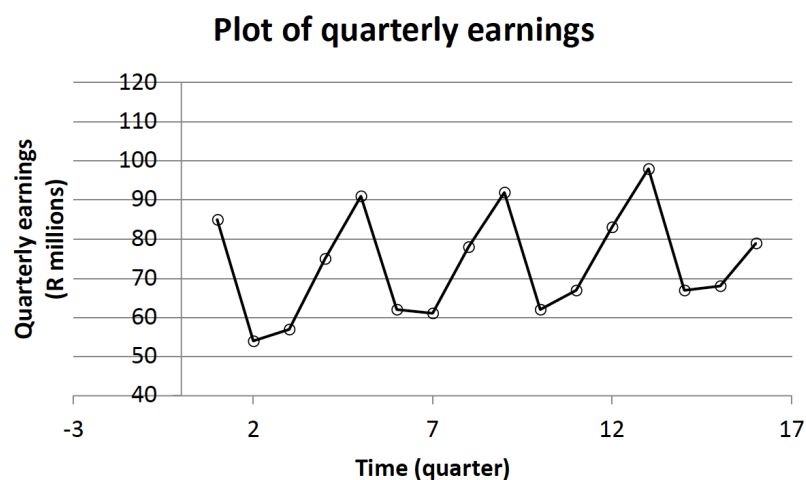
The quarterly earnings (in millions of Rands) of a cold drink manufacturer in SA are found in the table below.

Quarter	Year			
	2007	2008	2009	2010
1	85	91	92	98
2	54	62	62	67
3	57	61	67	68
4	75	78	83	79

Table: Quarterly earnings (in R millions)

	Standard Error			
	Coefficients		t Stat	P-value
Intercept	68.75	7.1419	9.6263	1.49E-07
Time	0.5809	0.7386	0.7865	0.4447

Table: Regression output



- What time series components are visible in the plotted series?
- Consider the regression output:
 - Write down the trend equation.
 - Calculate the unadjusted seasonal index for the third quarter.
 - Given that the unadjusted seasonal index for the first quarter was 1.2560, interpret this value.

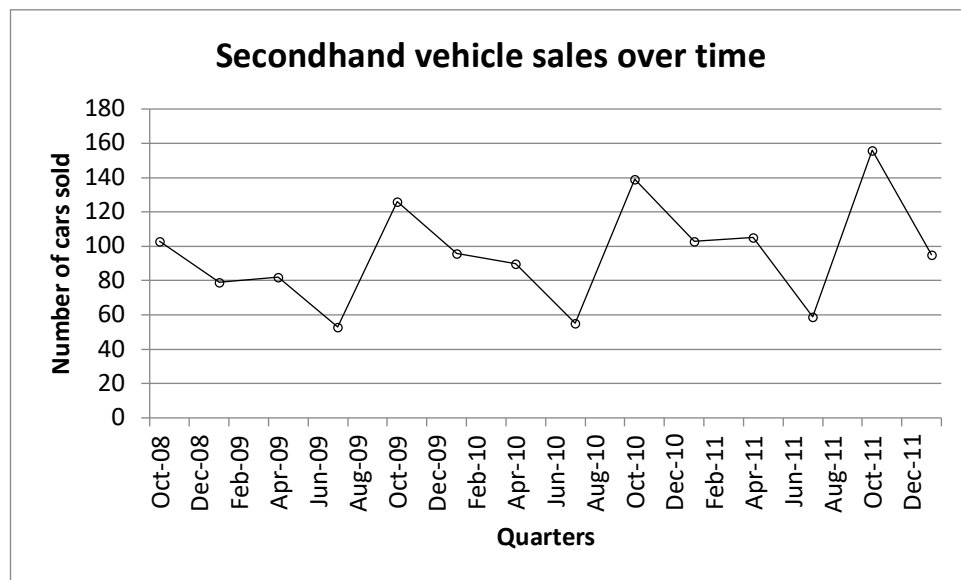
- c) Predict the earnings of the third quarter of 2011, using the trend equation and the seasonality index as appropriate.

Question 2

A used car salesman has recorded his quarterly sales figures, from the quarter ending October 2008 to the quarter ending January 2012 (refer to Table 6.1).

Year	2008	2009				2010				2011				2012
Month	Oct	Jan	Apr	Jul	Oct	Jan	Apr	Jul	Oct	Jan	Apr	Jul	Oct	Jan
Number of vehicles sold	103	79	82	53	126	96	90	55	139	103	105	59	156	95

Table 6.1: Used car sales figures



- a) Identify the time series components visible in the plotted time series.
- b) Assuming that the multiplicative decomposition method is appropriate to construct a forecast model, fill in the missing values from A to G in tables 6.2 and 6.3. Round all answers to **four** decimal places.

Time	Quarter	Sales	CMA(4)	Detrended data	Adj SI	Deseasonalised data
Oct-08	1	103			F	73.6264
Jan-09	2	79			1.0317	76.5732
Apr-09	3	82	82.125	0.9985	0.9892	82.8966
Jul-09	4	53	A	B	C	D
Oct-09	5	126	90.250	E	F	90.0672
Jan-10	6	96	91.500	1.0492	1.0317	93.0510
Apr-10	7	90	93.375	0.9639	0.9892	90.9841
Jul-10	8	55	95.875	0.5737	C	
Oct-10	9	139	98.625	1.4094	F	99.3599
Jan-11	10	103	101.000	1.0198	1.0317	99.8360
Apr-11	11	105	103.625	1.0133	0.9892	106.1481
Jul-11	12	59	104.750	0.5632	C	
Oct-11	13	156			F	111.5118
Jan-12	14	95			1.0317	92.0817

Table 6.2: Multiplicative decomposition model calculations

	2009	2010	2011	Unadjusted seasonal indices	Adjusted seasonal indices
Jan		1.0492	1.0198	1.0345	1.0317
Apr	0.9985	0.9639	1.0133	0.9919	0.9892
Jul	B	0.5737	0.5632	G	C
Oct	E	1.4094		1.4028	F
Sum:				4.0109	

Table 6.3: Seasonal index calculations

- c) The table (below) contains the regression output when the deseasonalised values were regressed on time (the quarter). Use the multiplicative decomposition model to predict the car sales for the quarter ended April 2013.

Regression output				
		Standard Error		
	Coefficient		t Stat	p-value
Intercept	77.176	3.315	23.280	2.35E-11
Time	2.129	0.389	5.467	0.0001